[SYNTHETIC PIPELINE SANITY TEST — NOT REAL LITERATURE]

Interferon-Gamma Receptor Ligation Triggers JAK1 Activation and STAT1 Expression in Macrophages

Authors: Synthetic Benchmark Generator
Provenance: Synthetic Pipeline Sanity Test (Procedural Fixture)

Page 1 — Abstract & Introduction

Interferon-gamma (IFN-gamma) orchestrates host antimicrobial defense and macrophage polarization toward the M1 phenotype.

Page 2 — Receptor Signaling

Ligation of IFN-gamma to its receptor induces trans-phosphorylation and activates JAK1 kinase at the cell surface.

Page 3 — STAT1 Phosphorylation

Active JAK1 phosphorylates STAT1 at Tyr701, driving STAT1 homodimerization and nuclear entry.

Page 4 — Gene Expression

Nuclear STAT1 binds GAS promoter elements to upregulate IRF1 expression in activated macrophages.

Page 5 — Discussion

In conclusion, IFN-gamma drives IRF1 expression via sequential JAK1 and STAT1 activation.